

# HARRIER – High Altitude Radiation Rate Ionisation Experiment Recorder

HARRIER is an experiment that is part of the New Horizons' Payload for launch at EUROOC 2026. It is a photodiode-based radiation detector that aims to measure the change in radiation rate at different altitudes. We expect the rate to increase with altitude as cosmic rays are less attenuated by the atmosphere at higher altitudes.

It consists of 2 separate boards; a detector board that contains the photodiode and outputs a digital signal and a power board that provides 25V and 5V voltages, as well as containing the on/off switch. They are stored along with their own separate battery in a 5mm thick casing that blocks alpha radiation.

## The experiment

HARRIER outputs a 'high' signal when a particle is detected. By monitoring the output channel, we store the timestamps where detection events occurred. This can allow us to synchronise the events with altitude readings on the Pickle-Rick's altimeter, producing a list of events and the altitudes they occur

## The detector

A photodiode is utilised to detect incoming radiation. When particles collide with the silicon surface of the diode, ionisation of donor atoms can occur, which results in additional free electrons being increased. This manifests as a small introduction of charge (on order of several femtocoulombs) and therefore current across the diode. To increase the sensitivity of the photodiode, a 25V reverse bias is applied this photodiode.

Then we convert this change in charge to a voltage output signal using a charge-sensitive preamplifier (CSP), which produces an output proportional to the integrated charge over its sampling time. This means small fluctuations in current can be detected once converted into a voltage output. During detection events, a voltage of ~7mV is expected to be produced. The CSP requires a 5V input.

This output is then fed into a comparator which will output a 'high' signal (5V) if a voltage threshold is exceeded during its sampling window. We set the voltage threshold with a voltage divider with a variable resistor. Otherwise, a 'low' signal of about 0V is outputted. We use this as an approximation of counts per second by counting how many times a high signal is outputted.

A voltage divider is used to reduce comparator output to 3.3V so it can be directly connected to a GPIO pin on the Pickle-Rick board, which will store the count rate data on an SD card.

Inputs required:

- +25V power rail for photodiode
- +5V power rail for comparator and CSP

Outputs:

- +3.3V digital output

Design considerations

We can also distinguish cosmic rays from EM radiation by their energy levels. Massive particles generally have far higher energies than photons from EM radiation. Therefore, by setting the threshold on the comparator above a certain level we can ignore low energy counts.

An issue with the detection of cosmic rays is that they are very fast (near light speed) so will not stay long inside the photodiode. By applying a 25V reverse bias voltage we can greatly increase the response time and sensitivity, allowing detection of these fast particles.

## Power board

The secondary board simply provides power for the main detector board. We must transform the battery power input into the correct voltages for the detector, which are 5V for the comparator and 25V for the amplifier.

The 25V input is provided by a buck boost from 7.4V to 27.5V, followed by an LDO regulator that reduces the voltage to 25V. This ensures a stable output signal as battery voltage is not consistent. For the 5V rail, we use another regulator to reduce the 7.4V input to 5V.

To turn the detector on/off, a MOSFET switch is used to switch the boost and 5V LDO on and off by disconnecting them from the battery input. This takes a 3.3V GPIO input from either the Flint and Steel board or the main Pickle Rick board. In turn, this deactivates the detector board. The power board should activate when the rest of the payload module turns on.

Inputs:

- ~7.4V battery input
- ~3.3V digital on/off input

Outputs:

- +25V rail
- +5V rail

Design considerations:

We decided to separate the detector board from the power board for 2 main reasons.

1. Horizontal spacing requirements in the payload module- the detector should roughly point upwards for the highest count rate of cosmic radiation, but this would make a combined board too wide to fit into the casing. By separating the boards, we can mount the detector horizontally and the power board vertically, ensuring we had enough space.
2. Testing and reliability - we could independently test the detector board and if necessary, replace it if it was not working or malfunctioned. This reduces the risks of failure as if either board fails, it can be replaced separately.

## Casing

The Harrier module is designed to be self-contained. It is connected to the Pickle Rick with the data output from the comparator and the on/off switch labels. It has its own separate battery within its case. We use a separate battery to ensure that power draw is more stable, so the input voltage and current is more consistent.

The casing consists of two 3D printed parts that slot together to form a closed box. Raised screws are used to hold the boards in place.

A key issue with photodiode detectors is that electromagnetic radiation will also be detected by the diode. We can avoid this by using a casing that blocks visible light from falling on the diode. Microwaves and radio waves may penetrate the casing, but they have far lower photon energies so are unlikely to cause any ionisation.